

The Formal Register

Every claim numbered, every dependency drawn and shipped, every obligation graded with the reason it stands: the corpus stated so that it can be checked rather than read.

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What this volume is for. The Monograph tells the framework as one continuous argument and is meant to be read. The Observational Standard holds it to the measured record. Systems of Interpretation records where it grips and where it slides off. This volume does none of those things. It states the framework as a formal object: one notation, numbered commitments, numbered definitions, numbered results with their hypotheses and their proofs or citations, a certified numerical register, the dependency graph with its complete edge ledger, and the proof obligations with the reason each is outstanding. A reader who wants to check whether the chain is sound should be able to do it here without reading the story at all.

What this volume is not. It is not a new derivation and it introduces no result that is not already carried by the Monograph or by a paper in the corpus. Where a proof is long it is stated compactly and the full derivation is cited by paper number. Where a claim is conditional the condition is written into the statement rather than into a footnote. Nothing here is graded higher than the corpus grades it elsewhere, and if this volume and any other document disagree, the Freeze Manifest of the current printing governs.

What changed in this printing. The dependency graph is now shipped as source and certified by a script rather than asserted, and its stated counts are corrected: the Fourth Printing reported thirty-one nodes and thirty-three edges, which were not reproducible because the generating source was never included. Four results are added, numbered T29 to T32, carrying Paper 21. Five of the seven proof obligations are regraded on the strength of them. The notation promise is bounded to what it actually delivers. Axiom six is reclassified as the commitment it is.

PART ONE

Notation

The symbols below are the ones the framework's own numbered claims use. This is a bounded list and it is stated as one: it does not claim to catalogue every symbol appearing anywhere in the corpus, because the papers carry working notation of their own that is defined where it is used. Earlier printings made the wider claim and did not deliver it.

phi	the self-description fixed point, $\phi^2 = \phi + 1$
r	the retained fraction of one act of self-description, $1/(2\phi)$
u	the outward fraction, $1 - r$
W_L W_B W_M	the three weights: direct, related, held
tau	an ordered operational traversal; also proper time where marked
[tau]	the equivalence class of traversals defining a recoverable object
Lambda	the carrier lattice; Λ^* its dual
Phi(E8)	the 240 roots of E8; $\alpha\beta\gamma$ individual roots
W(.)	the Weyl group of a root system
C	the system-preserving cluster projection (conditional expectation)
K	the decay generator; $B = \ker K$ the readable algebra
P_X P_T P_I	the geometry projectors: space, local clock, closure
V	the five-component cluster field
e, g	coframe and metric
D, F	covariant traversal operator and its curvature
G	the clustered gravity operator
M	the mass operator
Xi	the global carrier closure residue
lambda	the standing vacuum readout, $(4\pi / \sqrt{3}) r^{240}$
l	the carrier-to-physical length conversion
hbar	the quantum of phase

On identifiers. Claims are numbered A₁ to A₆, D₁ to D₁₈, T₁ to T₃₂ and O₁ to O₇, and those numbers are the corpus-wide identifiers used by every paper. They are not re-lettered in this printing. A longer prefixed form was considered and rejected: renaming would break the cross-references in nine issued papers to solve an ambiguity that does not exist, since no other numbering in the corpus uses these forms.

PART TWO

Commitments and admissibility conditions

These are what the framework commits to. Everything numbered in Part Four descends from them and from nothing else, which Part Six verifies. The heading says commitments rather than axioms because A₆ is not an axiom: it is the definition of a class of maps, and calling it an axiom misdescribed it in earlier printings.

- A₁** Systemhood. A system is a system: it may hold differences internally without ceasing to be one thing. AXIOM.
- A₂** The ground. The first distinction does not land in a waiting space; the holding of the distinction is what a where is. POSTULATE, stated in Movement Two rather than hidden.
- A₃** The two-sided reading. A self-description reads its mixed relation in both directions, outward against held and held against outward. POSTULATE. This single commitment fixes the factor of two in r , forces the carrier's self-duality, and forces its evenness. It is the join between the numerical branch and the carrier branch of the whole framework: see Part Six.
- A₄** Saturation. An accumulating single channel reaches a condition where one channel can no longer carry what it has taken on. POSTULATE. Where that condition sits is OPEN.
- A₅** Conservation of the readout. Nothing exposed at a closure event vanishes from the total of continuing face, outgoing record, and the ordering relation between them. POSTULATE.
- A₆** Admissible reduction. A system-preserving reduction is unital, idempotent, completely positive, bimodular over the readable algebra, and covariant under equivalent carrier descriptions. DEFINITION, not a choice among options: any map failing one of these five conditions deletes rather than decays. The

covariance clause is load-bearing beyond its appearance and is the single condition Paper 21 leans on; see O1.

PART THREE

Definitions

- D1** Distinction. That which prevents complete sameness. Primitive.
- D2** Identity. The recoverability of a distinction through admissible change.
- D3** Relation. The operation through which distinct contents affect, compare or transform one another without being erased.
- D4** The seven doings. Distinguish, hold, close, answer, change, order, record. The constitutive grammar.
- D5** Traversal stages. Persist, distinguish, relate, propagate, resolve, return, close. The grammar of one act passing through a world of finished things. Related to D4 by the noun rule: the first stage is the first three doings completed.
- D6** Object. A stable equivalence class of closed traversals. Objecthood is retrospective.
- D7** The update. The two-channel step carrying (resolved, held) to (resolved plus held, resolved).
- D8** Channels. The outward share u and the retained share r , with u plus r equal to one.
- D9** The three weights. The terms of $(u + r)$ squared: W_L equals u squared, W_B equals $2ur$, W_M equals r squared.
- D10** Crossing. One act of self-description enacted across a channel. Its price is the retained fraction r of T_4 ; see T6 for why the price is exact.
- D11** Record. A distinction that remains discriminable and recoverable under the relevant future dynamics. Two-valued: there is no partially recoverable record.
- D12** Observer. Any system in which a distinction of another system becomes physically related and persistently retained. Mechanical observer: an observer with staged architecture, being a prepared ready state, a channel-defining coupling, propagation, resolution, amplification, and record production.
- D13** Carrier. The structure on which a self-describing record is written, required to equal its own dual.
- D14** Primitive crossing. The smallest nonzero admissible carrier displacement. In E8 these are exactly the roots, of squared norm two.
- D15** Cluster projection. The reduction C from carrier observables onto the readable subalgebra, satisfying A6. Its concrete form is T_{31} .
- D16** Decay generator. K , the difference between doing nothing and applying C . Minimal form I minus C . See T14 for why the minimal form is not always the one meant.
- D17** Readable algebra. B , the part of the carrier algebra the decay generator leaves alone. This is the physical theory.
- D18** Geometry projectors. Mutually orthogonal P_X , P_T , P_I onto external distinction, local causal order and system closure, summing to the geometry block. The physical projector is P_X plus P_T ; P_I is retained as closure data and excluded from traversable directions.

PART FOUR

Results

Each result carries its hypotheses, its grade, and either a compact proof or a citation to the paper that carries the full derivation. A result stated as **CONDITIONAL** names its condition in the statement itself. T29 to T32 are new in this printing and are carried by Paper 21.

- T1** The seven doings admit a unique linear order. **PROVED**. Nothing is held before it is drawn, nothing closes before it is held, nothing answers before there is a closed thing to answer, nothing changes except by answering, nothing orders except as change follows change, nothing is recorded before it has happened. The dependency relation is a strict partial order with a unique linear extension. Monograph, Movement Two.
- T2** The minimal update is unique. **PROVED**. Requiring an update that adds and never subtracts, carries exactly one clean copy of each channel forward, and promotes the present resolved amount to the next reserve, admits exactly one operator. Movement Four.
- T3** The stable ratio is the golden fixed point. **PROVED**. Requiring the update to preserve the ratio between channels while changing overall size yields $\phi^2 = \phi + 1$, which has exactly one positive solution. Movement Four.
- T4** The retained fraction is $1/(2\phi)$. **PROVED** given A3. The two-sided reading measures the retained side with a factor of two; u and r are then pinned with no adjustable quantity. Movement Four.
- T5** The three weights exhaust the whole. **PROVED**. Two channels squared give four combinations; the two mixed terms are the same pair read in opposite order and count once when reading direction is irrelevant. The three terms sum to one. Movement Five.
- T6** The quantised crossing. **PROVED** given D11. A crossing either closes, at price exactly r , or fails to traverse. Intermediate prices would require a partial closure, and D11 admits none. Exactness follows from the discreteness of closure rather than from tuning. Paper 18.
- T7** The traversal price in closed form. **PROVED**. $r^3 = 1/(8\phi^3)$, all over eight. Movement Seven; the carrier derivation of the exponent is T10.
- T8** The carrier is E8. **CONDITIONAL** on A3. Positive-definite even unimodular lattices exist only in ranks divisible by eight; at rank eight there is exactly one, namely E8, with 240 roots. At rank sixteen there are two and at rank twenty-four there are twenty-four, so the uniqueness is a property of the first realised rank. Movement Nine; lattice classification borrowed with its name attached.
- T9** Evenness descends from the two-sided reading. **PROVED** given A3. A relation read outward and back counts its self-pairing twice, so carrier norms fall on the even integers. Positive-definiteness follows separately from D1, since an element pairing to zero with itself is a distinction making no difference to itself. This reduces T8 from three stipulations to one. Movement Nine.
- T10** The minimal closed carrier history is triangular. **PROVED**. For roots α and β with inner product minus one, $\alpha + \beta$ is a root; setting γ to its negative gives a sum of zero. Immediate reversal is cancellation and encloses nothing, so three is minimal. With T6 the weight of that history is r^3 . Paper 20.
- T11** The root cell area is root three. **PROVED**. The parallelogram spanned by two roots at that angle has area the square root of four minus one; the closed triangle is half of it. This identifies the root three carried since Movement Nine as a covolume. Paper 20.
- T12** The golden ring pairing. **PROVED** numerically to machine precision. Projected onto the Coxeter plane the 240 roots fall into eight rings of thirty whose radii pair off, four pairs, each in the ratio ϕ , with residual

below one part in ten to the fifteenth. Research Archive; reproducible from the Cartan matrix and the eigenvectors of the Coxeter element.

- T13** The physical algebra is the kernel of the decay. PROVED given A6. System identity cannot decay, so it lies in the kernel; what the generator suppresses becomes unreadable; the readable algebra is exactly the fixed algebra of the reduction. Paper 20; Movement Ten.
- T14** Minimal and general decay generators are distinct. PROVED. The minimal generator I minus C is idempotent, hence has spectrum in the two-point set, and its adjoint action has eigenvalues in minus one, zero and one. That suffices for T13 and for the closure theorem, and it does not suffice for a graded descent or for a mass spectrum, which would collapse to two values. The physical generator is any positive operator annihilating the identity whose kernel is B . Paper 20, Appendix.
- T15** The local clock mode is single. PROVED. The record accumulates along one monotone order; two independent orders of accumulation would be two records and therefore two systems. Movement Eleven, from the arrow of Movement Thirteen.
- T16** The closure mode is single. PROVED. There is one enacted closure whose record is inhabited; the closure mode counts enacted closures. Movement Eleven, from the one-world result of Movement Twenty-Three.
- T17** The inherited form is Lorentzian. CONDITIONAL on the opposite closure roles of spatial opening and causal completion. The inherited quadratic form is P_X minus P_T , up to overall sign and the conversion factor, which is the unique minimal nondegenerate form admitting null directions. Movement Twelve.
- T18** Second witness for T17. PROVED, cited. Any bijection preserving the causal order of a flat continuum of more than two dimensions is a Lorentz transformation composed with a translation and a dilation, the dilation being the unit convention. Alexandrov and Zeeman; Paper 19. Independent of T17's route and reaching the same signature.
- T19** Superposition is forced. PROVED. If two actions are admissible and no physical event has distinguished one, selecting either adds information the system does not contain. The description must retain both. Movement Seventeen.
- T20** Complex phase is forced. PROVED. Additive action under sequential composition, unit modulus under reversible carriage, and continuity admit exactly one family of weights, the rotations, giving the exponential of the action in units of the quantum of phase. Movement Seventeen.
- T21** The Born pricing is forced. PROVED, cited. Requiring the same physical effect to carry the same probability in every complete measurement context admits exactly one pricing, the squared magnitude. Gleason, with the extensions of Busch and of Caves, Fuchs, Manne and Renes for the low-dimensional cases. Movement Eighteen.
- T22** Closed carriage is unitary. PROVED, cited. A transformation preserving every transition probability is unitary or antiunitary; continuity with the identity excludes the second. Its generator exists and is self-adjoint. Wigner; Stone. Movement Eighteen.
- T23** The boundary weight is the interference term. PROVED as arithmetic. Expanding the squared magnitude of a sum of two unresolved alternatives at primitive magnitudes u and r gives 0.477457514, 0.427050983 and 0.095491503, summing to one: exactly W_L , W_B and W_M . The cross-term, belonging to neither alternative alone, is interference. CONDITIONAL as physics on the primitive magnitudes being u and r . Movement Eighteen.
- T24** The three weights structure the gravitational account. PROVED as algebra. The square of a defect formed from curvature weighted by u against realised area weighted by r has three terms and no others: the closed curvature record, the curvature-realisation cross-term, and the realised volume. Local gravitational dynamics reside in the cross-term. PROPOSED as physics for the identification of that cross-term with ordinary gravitational dynamics. Movement Nineteen; Paper 20.

- T25** Sector classification by one operator. PROVED given T14. Carrier operations commuting with the decay preserve the clustered identity and are gauge; operations connecting distinct decay sectors are matter when they close into a cycle reproducing their readable identity; failure to commute measures the cost of persistence and is mass. Modes commuting with the decay are massless. Movement Twenty; Paper 20.
- T26** Retention over replacement. PROVED given D7. A persistent self-description must join each new resolution to the record rather than overwrite it, or its later states are not records of the same system. Movement Twenty-Six.
- T27** Records are single, and which record is unreadable. PROVED. A state in which both alternatives remain coherently present is not discriminable as either and is therefore not a record; singleness is the definition of one. Any internal answer to which outcome would itself be a record produced by a resolving event, so the question regresses and the selection is not readable from within the record domain. Paper 18.
- T28** No recollapse while the record stands. CONDITIONAL on A5. Contraction of the causal domain toward coincidence would press resolved distinctions back into unread simultaneity, which is overwrite by geometry and is forbidden by T26. Paper 18.
- T29** The spatial rank is the dimension of the surviving block. PROVED given T15, T16 and D18. Since the geometry projectors are mutually orthogonal and the clock and closure modes are each single, rank P_X equals three is identical to the statement that the geometry block has dimension five, and therefore, in a rank-eight carrier, to the statement that the decayed part has rank three. Paper 21, Section Two. New in this printing.
- T30** E8 has exactly three rank-three subsystems. PROVED. Up to Weyl conjugacy they are $A_1+A_1+A_1$, A_2+A_1 and A_3 , with orthogonal complements D_4+A_1 , A_5 and D_5 respectively. The enumeration is exhaustive because the Weyl group is transitive on the 240 roots, so every rank-three subsystem is conjugate to one containing a fixed root. All three leave a rank-five survivor. Paper 21, Section Three. New in this printing.
- T31** The decay is on A_3 and the surviving symmetry is D_5 . CONDITIONAL on the reduction averaging over a reflection subgroup of the carrier, which is the reading of A_6 's covariance clause recorded as O_1 . Given that, $A_1+A_1+A_1$ is excluded by T10 for containing no closed triangular history, and A_2+A_1 is excluded by T25 for being reducible and therefore two decay operators rather than one. The survivor is unique. Its reduction is averaging over $W(A_3)$, the symmetric group on four letters; its gauge algebra is $so(10)$; its matter modules are the spinor sixteen and the vector ten; and its family index splits as one invariant direction plus three, giving three families under T25's criterion. Paper 21, Sections Four to Seven. New in this printing.
- T32** A complete carrier closure exists. PROVED by explicit construction. The 240 roots sum to zero because a root system contains the negative of each root, so any traversal using each root once closes automatically. An ordering in which every step is itself a root exists, and a 240-step witness is recorded in the certificates and verified independently of the search that produced it. Uniqueness and the number of such closures remain OPEN and are not required for the vacuum readout, since every complete closure has exactly 240 crossings. Paper 21, Section Eight. New in this printing.

PART FIVE

The numerical register

Every constant the framework uses, in closed form and to twelve significant figures. No value on this list is fitted, and no value is obtained by comparison with a measurement. All are reproducible from the Reproducibility layer by a single command.

ϕ	$(1 + \sqrt{5})/2$	1.61803398875
r	$1/(2\phi) = (\sqrt{5}-1)/4$	0.309016994375

u	1 - r	0.690983005625
W_L	u^2	0.477457514063
W_B	2ur	0.427050983125
W_M	r^2	0.0954915028125
	W_L + W_B + W_M	1.000000000000
r^3	(sqrt5 - 2)/8	0.0295084971875
1 - r^3	retained, one crossing	0.970491502813
(1-r^3)^3	retained, three crossings	0.914061068091
sqrt3	root cell parallelogram	1.73205080757
sqrt3 / 2	closed triangle cell	0.866025403784
r^240	global carrier residue	3.94246077122e-123
lambda	(4 pi / sqrt3) r^240	2.86033313614e-122

Three of these deserve a note. The traversal price is written as a surd throughout the corpus because a decimal in that position reads as something fitted, and the closed form shows what it is: three copies of one retained fraction, multiplied because the channels are crossed in series. The global residue is one retention per root over a complete closure, and as of this printing that closure is proved to exist rather than granted, which is T32. The standing vacuum readout is a computed value with a stated construction and no adjustable quantity anywhere in its map; Paper 21 Section Nine reports its agreement with the measured cosmological constant, and that agreement is offered as a test the framework passes rather than as a derivation of the measurement.

PART SIX

The dependency graph

The Monograph's governing promise is that nothing is slipped in by hand. That promise is a claim about structure and can therefore be checked rather than believed. Draw every commitment, definition, result and obligation as a node, draw an edge from each to everything it uses, and ask whether the graph contains a cycle. A cycle would mean a claim standing on itself, which is the formal shape of the failure the promise forbids.

This printing ships the graph as source. Every edge below carries the phrase from this volume's own wording that licenses it, so a reader can check each one against the text rather than taking the count on trust. The Fourth Printing stated thirty-one nodes and thirty-three edges; those figures were not reproducible, because the generating source was never included, and they are corrected here.

64 nodes, 67 dependency edges, no cycles

Two kinds of link are deliberately excluded from the graph and both exclusions are recorded in the source. A reference is a pointer for the reader, not a dependency: D10 points at T6 and D16 points at T14, and treating either as a dependency manufactures a false cycle, since T6 depends on D10 and T14 depends on D16. A witness is an independent corroboration rather than a step in the chain: T18 reaches T17's conclusion by a separate route and is not part of the derivation of it.

The load-bearing commitments, those with something depending on them, are A3, A5, A6, D1, D10, D11, D13, D14, D15, D16, D17, D18, D4, D7, D9. The remaining nine, A1, A2, A4, D12, D2, D3, D5, D6, D8, are stated but carry nothing in the numbered chain. That is not automatically a fault, since several are framing rather than machinery, but it is recorded here rather than left to be discovered. A6 and A3 carry the most.

The longest chain in the corpus is 7 steps. There is no deep sequence in which an error could hide: the structure is shallow and wide, which is what a framework that unfolds from a seed looks like, and it is the checkable form of the claim that the whole thing comes from one line.

The edge ledger

Read each line as: the claim on the left uses the claim on the right, for the reason quoted.

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D10 <- T4    Its price is the retained fraction r
T1  <- D4    The seven doings admit a unique linear order
T2  <- D7    The minimal update is unique
T3  <- T2    Requiring the update to preserve the ratio between channels
T4  <- A3    PROVED given A3
T4  <- T3    u and r are then pinned
T5  <- D9    the terms of (u + r) squared
T5  <- T4    Two channels squared give four combinations
T6  <- D10   A crossing either closes, at price exactly r
T6  <- D11   PROVED given D11
T7  <- T4    r cubed equals one over eight phi cubed
T7  <- T6    three copies of one retained fraction
T7  <- T10   the carrier derivation of the exponent is T10
T8  <- A3    CONDITIONAL on A3
T8  <- D13   required to equal its own dual
T8  <- T9    This reduces T8 from three stipulations to one
T9  <- A3    PROVED given A3
T9  <- D1    an element pairing to zero with itself is a distinction making no
T10 <- D14   In E8 these are exactly the roots
T10 <- T6    With T6 the weight of that history is r cubed
T10 <- T8    For roots alpha and beta
T11 <- T10   the closed triangle is half of it
T12 <- T8    Projected onto the Coxeter plane the 240 roots
T13 <- A6    PROVED given A6
T13 <- D15   the fixed algebra of the reduction
T13 <- D17   the readable algebra is exactly the fixed algebra
T14 <- A6    the definition, not a choice among options
T14 <- D16   The minimal generator I minus C is idempotent
T15 <- D11   two independent orders of accumulation would be two records
T15 <- T26   from the arrow of Movement Thirteen
T16 <- T27   from the one-world result of Movement Twenty-Three
T17 <- D18   The inherited quadratic form is P_X minus P_T
T17 <- T15   the opposite closure roles of spatial opening and causal completio
T17 <- T16   the opposite closure roles of spatial opening and causal completio
T19 <- D11   no physical event has distinguished one
T20 <- T19   Additive action under sequential composition
T21 <- T20   the same physical effect to carry the same probability
T22 <- T20   A transformation preserving every transition probability
T23 <- T5    exactly W_L, W_B and W_M
T23 <- T21   Expanding the squared magnitude of a sum
T24 <- T5    has three terms and no others
T25 <- T14   PROVED given T14
T26 <- D7    PROVED given D7
T27 <- D11   is not discriminable as either and is therefore not a record
T28 <- A5    CONDITIONAL on A5
T28 <- T26   which is overwrite by geometry and is forbidden by T26
T29 <- D18   P_X, P_T, P_I sum to the geometry block
T29 <- T15   rank P_T = 1
T29 <- T16   rank P_I = 1
T30 <- T8    rank-3 root subsystems of E8
T31 <- A6    the reduction averages over a reflection subgroup
T31 <- T10   A1+A1+A1 hosts no closed triangular history

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T31 <- T25    A2+A1 is reducible, which is two operators not one
T31 <- T29    a rank-5 survivor means a rank-3 decayed part
T31 <- T30    exactly three candidates
T32 <- D14    every step is a primitive crossing
T32 <- T8     the 240 roots
O1  <- D15    A concrete representation of the cluster projection C
O2  <- T17    the surviving local symmetry must be the one T17 derives
O2  <- O1     OPEN, conditional on O1
O3  <- O1     OPEN, conditional on O1
O4  <- D14    a complete carrier closure
O4  <- T8     passes each of the 240 roots exactly once
O5  <- D11    the growth of the record
O7  <- T6     quantised retention at powers of r
N_lambda <- T8    (4 pi / sqrt3) r^240
N_lambda <- O4    The global vacuum residue rests on it

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The graph is acyclic. This is the strongest structural statement the corpus can make about itself, and it is worth being exact about what it does and does not establish. It establishes that no claim in the framework is used in its own derivation. It does not establish that the claims are true, and it does not establish that the roots are the right roots. It rules out one specific failure, circularity, completely and checkably. The remaining ways to be wrong are in Part Seven and in Paper 21 Section Twelve.

PART SEVEN

Proof obligations

What the framework owes, with the reason each item stands. The corpus does not object to open questions; it objects to questions left open without anyone having looked. Every item below has been attacked, and five of the seven moved in this printing.

Before the list, one structural fact, because it changes how the list should be read. No result from T1 to T32 depends on any obligation. The obligations are terminal: nothing in the argument hangs off them. The single thing in the corpus resting on an open item is the standing vacuum readout, which rested on O4 and no longer does, since T32 proves the closure exists. The argument is therefore closed, and what follows is a forward work programme rather than a list of holes. That is checkable from the edge ledger in Part Six and is certified by the graph script.

- O1** That the admissible reduction of A6 is an average over a reflection subgroup of the carrier. This is the reading of A6's covariance clause that Paper 21 leans on, and it has replaced four separate obligations with one. A conditional expectation covariant under equivalent carrier descriptions is an average over the group relating them, and the resulting map satisfies all five conditions of A6 without adjustment, which is evidence but not proof. OPEN, and now the single load-bearing question in the representational half of the corpus. If it falls, T31 falls, and with it the spatial rank, the gauge algebra, the matter modules and the family count, together.
- O2** That the surviving spatial modes number three. CLOSED conditionally by T29, T30 and T31. Given O1 the answer is forced and unique, and it is the same fact as the surviving symmetry being $so(10)$. This item no longer waits on a representation; it determines one.
- O3** The low-energy gauge algebra, graded matter modules, chirality, families and anomaly freedom. Substantially answered by T31: gauge $so(10)$, matter in the spinor sixteen with a vector ten, and three families from the one plus three splitting of the family index under the decay's own Weyl group. What remains OPEN is chirality, meaning the mechanism that makes the three surviving directions left-handed in the observed sense, and anomaly freedom, which is a calculation in the named modules rather than an open form.

- O4** That a complete carrier closure passes each of the 240 roots exactly once. CLOSED for existence by T32, with an explicit witness. What remains OPEN is uniqueness and the number of such closures, and neither is required by anything the corpus claims.
- O5** The integral joining the growth of the record to the measured expansion history. The zeroth-order clause $H \times t$ equals one is PROVED in Paper 19. The residual is the integrated traversal price, and Paper 21 Section Nine shows that the standing vacuum readout, now resting on a proved closure, implies a dark energy fraction and an age of the universe that bracket the measured values with nothing fitted. OPEN as a derivation of the residual from first principles; the consistency check is passed and is not the same thing.
- O6** The audited closure counts of the two measurement pipelines that disagree about the expansion rate. OPEN and empirical, unchanged. The framework's declared count is three, its prediction is exposed rather than supported, and the arithmetic now ships: the stake stands at a pull of 0.55 standard deviations. The audit itself, from pipeline architecture rather than from the size of the gap, has not been done.
- O7** A running, self-holding system returning quantised retention at powers of r under a convention declared before the count. OPEN and empirical. This is the framework's own stated condition of defeat and cannot be settled at a desk. It is not regraded and should not be.

One reading, one chirality argument, one anomaly calculation, one closure count, and two measurements. That is the complete list, and it is shorter than the list this printing inherited because four items collapsed into one when the ledger was attacked from the other end. Everything else in the chain from A1 to T32 is proved, conditional on a condition written into its own statement, or dissolved as a unit convention.

PART EIGHT

Where each result lives

Every numbered result, and the paper or movement carrying its full derivation.

T1 T2 T3 T4 T5	Monograph, Movements Two, Four and Five
T6 T27 T28	Paper 18, The Quantised Crossing
T7	Monograph, Movement Seven; exponent from T10
T8 T9	Monograph, Movement Nine; Papers 08 and 09
T10 T11	Paper 20, The Forward Closure, Section Four
T12	Reproducibility, e8_golden_tests.py; Paper 09
T13 T14	Paper 20, Sections Five and Appendix; Movement Ten
T15 T16	Monograph, Movement Eleven
T17	Monograph, Movement Twelve; Papers 16 and 17
T18	Paper 19, The Closing Ledger, Section Two
T19 T20	Monograph, Movement Seventeen; Paper 13
T21 T22 T23	Monograph, Movement Eighteen; Papers 13 and 15
T24	Monograph, Movement Nineteen; Paper 20, Section Nine
T25	Monograph, Movement Twenty; Paper 20, Section Ten
T26	Monograph, Movement Twenty-Six
T29 T30 T31 T32	Paper 21, The Rank-Three Decay
Papers 04 05	the realisation principle and the snowflake closure theorem
Papers 08 09 10	the self-dual carrier, the golden skeleton, the root three
Papers 11 12	the interface closure and its proof ledger; historical
Papers 13 to 17	the semantic layer and the logic of physical action
Papers 18 19	the exact crossing price and the closing ledger
Papers 20 21	the forward closure and the rank-three decay

The Monograph carries the argument as a story and grades every claim in place. The Observational Standard carries what may honestly be said about a measured system. Systems of Interpretation carries

the record of the framework being thrown at real systems, including its misses, which fall in the regimes its own grammar forbids, and which are retained deliberately: a framework that reported only its successes would be worth less. The Reproducibility layer carries the data, scripts and certificates, and every number in Part Five and every result in Paper 21 regenerates from it by one command. This volume carries none of that and duplicates none of it. It carries the shape.

PART NINE

Concordance of movement numbers

The Fourth Printing inserted four movements into the Monograph, so every movement from the tenth onward carries a number one to four higher than it did in the Third. Papers 18 and 19 were issued against the earlier numbering and are preserved as issued rather than silently edited, in keeping with the corpus discipline that corrections belong in new printings rather than in altered files. A reader following a cross-reference from either of those papers should read it through this table. Papers 20 and 21 use the current numbering throughout, and the Fifth Printing changes no movement numbers.

Third	Fourth/Fifth	Movement	
-	10	The descent	(new in Fourth)
-	11	What survives	(new in Fourth)
-	12	The shape of the world	(new in Fourth)
10	13	The record of the whole	
11	14	The equation as a finished journey	
12	15	The action and its double	
13	16	The container	
14	17	The unresolved world	
15	18	The bookkeeping of chance	
16	19	The carrying of sameness	
-	20	The one operator	(new in Fourth)
17	21	The extended traversal	
18	22	The watcher's machinery	
19	23	The ocean	
20	24	The architecture of one world	
21	25	The luminous wound	
22	26	The wound that could not forget	
23	27	What it claims, and what it leaves open	